

OpenVAF — Verilog-A LRM Compliance

Clause-by-clause coverage of the Verilog-A language (VAMS LRM 2023, Annex C)

Ngspice + OpenVAF Enhancements project

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OpenVAF — Verilog-A LRM Compliance

How the `openvaf-r` compiler in this repository covers the Verilog-A language, clause by clause against the Verilog-AMS Language Reference Manual (the 2023 edition, [docs/VAMS-LRM-2023.pdf](#); Verilog-A is its Annex C analog-only subset). Every claim in this document is backed by a committed, executable check: the code examples are drawn from the verify suites under [examples/](#), which compile each construct with the committed compiler, simulate it with the committed ngspice, and compare the numbers against closed-form expectations on every regression run.

Status marks used throughout:

Mark	Meaning
✓	fully implemented and runtime-verified
⚠	implemented with LRM-sanctioned fallback semantics, or a documented deviation
×	out of scope: a Verilog-AMS (mixed-signal) feature outside Annex C

A one-page summary table closes the document. Companion references: the [User Handbook](#) (task-oriented), the [openvaf change report](#) (where each capability was implemented and why), and the per-enhancement write-ups in [enhancements_doc/](#).

1. Compilation model and scope (LRM 1, Annex C)

OpenVAF compiles one Verilog-A source file (with its ``include` closure) into an OSDI shared object: analytic residuals and Jacobians are generated by automatic differentiation, so every construct below is “supported” only if it produces correct derivatives as well as correct values — a stricter bar than parsing. Hierarchy is resolved by a compile-time elaboration pass (§7), after which the pipeline sees a single flat module.

Annex C draws the language boundary this project targets: the analog subset. Digital and mixed-signal constructs (reg/wire logic, digital `always/initial`, connect modules, ``default_discipline`, discrete-domain disciplines) are × out of scope by definition, and the compiler rejects them with located diagnostics rather than accepting them silently — e.g. `domain discrete` on a discipline with natures is a proper two-label error citing LRM 3.6.2.2.

2. Lexical conventions (LRM 2) — ✓

Identifiers, including escaped identifiers (LRM 2.7.1):

```
electrical \node-1a ;           // any printable characters until whitespace
parameter real \my.gain = 2.0;
```

Numbers (LRM 2.5): decimal integers, reals with exponents, SI scale suffixes (1k, 2.5u, 10n), and based integer literals in all four bases with optional size and sign, underscores as separators:

```
integer a, b, c, d;
analog begin
  a = 8'hFF;           // sized hex           -> 255
  b = 'sb1010_1100;    // signed binary with separators
  c = 'o17;            // octal               -> 15
  d = 4'd12;           // sized decimal
end
```

The don't-care digits x/z/? are additionally accepted in the power-of-two bases when the literal is a casex/casez item (§5.4); anywhere else they are a located error rather than silent zeros.

The three-token forms of LRM 2.6.1 work as of E-515: the digits may be separated from the base by white space and each token may come from a macro — 5 'D 3 (the LRM's own Example 2), 12 'b 0011_0101_0001, `SZ'hFF, 'h 837FF, 8'sh FF — evaluated in the `lrmlex` suite next to the contiguous forms. A spaced literal whose digits are illegal for its base is a located error. Two illegal forms that used to compile in silence are errors now: 1. (LRM 2.6.2 requires a digit on each side of the decimal point) and a string spanning a raw newline (LRM 2.7). A backslash immediately before the newline is a line continuation instead: the backslash-newline pair contributes nothing to the string (SystemVerilog 5.9 semantics and de-facto Verilog-A practice — BSIM4 splits its long `$strobe` messages exactly this way), so only an unescaped newline is refused. `assert`, `root` and `do` — none of them reserved by Annex B — are legal identifiers again (`do` contextually, so `do-while` still parses), and attribute instances parse in the expression positions of LRM 2.9, with duplicate attributes resolving to the last value as required.

Strings (LRM 2.6.3) with the full escape set — `\n`, `\t`, `\\`, `\"`, and octal `\ddd` — processed by a single-pass unescaper (a sequential-replacement implementation corrupted overlapping escapes and was rewritten). Comments both styles, including the edge case of a `//` comment at end-of-file without a trailing newline (formerly a lexer hang). Attributes (`* ... *`) parse everywhere the LRM allows and carry simulator-facing metadata (§9.6).

3. Data types (LRM 3)

3.1 Value types — ✓

integer (32-bit), real (IEEE double), and string variables, with genuine persistence across evaluations for both numeric types (a Verilog-A module variable keeps its value between solver calls, the foundation of event counters and iterative models):

```
integer count;           // persists across evaluations
analog begin
    @(cross(V(in) - 0.5, +1))
        count = count + 1; // genuinely accumulates
end
```

Uninitialized strings default to `""`; declaration initializers work on scalars and (multi-dimensional) arrays, evaluating once at setup when parameter-only:

```
real x = 2.0 * gain;
real taps[0:3] = '{0.1, 0.2, 0.3, 0.4};
```

A string literal assigned to an integer converts per LRM 3.3: the characters are treated as a packed byte vector, right-justified and truncated on the left / zero-filled to 32 bits ("A" is 65, "AB" is 16706 = 0x4142, "ABCDE" keeps its last four bytes). Only the *literal* converts — a string value still cannot be assigned to an integral type, exactly as the LRM words it.

3.2 Parameters (LRM 3.4) — ✓

Ranges and exclusions enforced on user-given values; `localparam` genuinely non-overridable; `alias-param` including `$param_given` through the alias — with both of LRM 3.4.7's error rules enforced: naming the alias and its target on one `.model/instance` line is an error (in the `ngspice` netlist parser and in the OSDI `setup_model` path both), and referencing the alias anywhere in the module body is a targeted compile error; string parameters (usable in case dispatch); array parameters of any dimensionality, expanded to per-element OSDI parameters that are individually overridable from SPICE and overridable whole at instantiation (`leaf #(.w( '{1,2,3})) l1(a,b);`, with the element count checked against the declaration):

```

parameter real r = 1k from (0:inf) exclude 1e6;
parameter integer sel = 0 from [0:15];
localparam real g = 1.0 / r;           // tracks r, not overridable
parameter real w[0:1][0:2] = {'{1,2,3}', '{4,5,6'}};
// SPICE: .model m dev(w[1][2]=9.5)

```

Two typing edges are deliberate deviations, documented in the [handbook's limitations chapter](#): an untyped parameter's type is frozen at compile time from its default (parameter untyped = 1; is an integer parameter, and a real override is rounded — with an ngspice warning — where LRM 3.4.1 would re-type the parameter from the final overridden value; a compiled OSDI descriptor declares exactly one type per parameter). $\triangle$ And in the lenient direction, an untyped string or array parameter is accepted with its type inferred from the default, although LRM 3.4.1 makes the type mandatory for those two kinds.

One deliberate reading of the LRM, matching industry practice (the CMC convention): a parameter's default is exempt from its own range — compact models use out-of-range defaults as "feature disabled" markers — while every user-given value is fully validated. $\triangle$ documented in the [handbook's limitations chapter](#).

Block-scoped parameters (LRM 6.3): a parameter/localparam declared inside a named begin: label block is a compile-time constant local to that block, read hierarchically and derivable from the module's parameters (E-87):

```

analog begin
  begin: s
    parameter real g2 = gain * gain; // block param, from a module param
    real contrib = g2 + 1.0;
  end
  vout = s.contrib;                  // hierarchical read
end

```

Overriding a module parameter propagates into the block-scoped parameters that depend on it. The LRM's `#(.blk.p(4))` instance override of a block-scoped parameter is illegal (LRM 6.3.2) and rejected with a targeted diagnostic.

paramset blocks (a Verilog-AMS 6.4 feature adopted because real model libraries use them) work, including hidden-system-parameter assignments:

```

paramset fast_nmos nmos_va;
  .w = 2e-6; .l = 60e-9;
  .$mfactor = 4;
endparamset

```

3.3 Natures, disciplines, nets (LRM 3.5–3.7)

✓ Discipline/nature declarations with attribute access from expressions (`net.potential.abstol`); derived natures — nature `Xf` : Flow and deriving from a discipline's nature (`: electrical.flow`) — with full attribute inheritance; ground declarations in both orderings; vectored nets and ports with bit-selects; net initializers that become solver nodesets; domain continuous:

```

nature CurrentX : Current; // derived nature, inherits abstol/units
  abstol = 1e-15;
endnature

electrical [3:0] bus;        // vectored net (range-then-name)
electrical data[0:7];        // vectored net (name-then-range, E-89)
electrical vout = 2.5;       // initializer -> OSDI nodeset (initial guess)
ground electrical gnd;

```

Both declaration orders are accepted for vectored nets and ports: the range-then-name form `electrical [3:0] bus; / input [3:0] bus;` and the name-then-range form `electrical bus[3:0]; / input bus[3:0];` (E-89). A bit of a multi-bit input bus port reads its own terminal regardless of the port's position in the header — a bus port that is not the last port keeps its bits contiguous and in header order, so the simulator's positional terminal mapping stays correct (E-90). The name-then-range form also accepts a multi-name list with per-name widths (`input a[0:1], b[0:3], c;`, E-91). A net/port/array width may reference a parameter (`electrical [0:N-1] out;`, `real w[0:N-1];`, E-91): it is folded from the parameter's elaboration-time default and fixed at that value — a structural parameter, since the OSDI descriptor has one node count per module. Such a parameter is frozen to a **localparam** (E-92) so a netlist override cannot resize the compiled structure (or desync it from behavioural code); to change the width, edit the default and recompile. Any **localparam** (frozen width parameter or hand-written) is flagged non-settable in the OSDI descriptor, so ngspice warns rather than silently ignoring a netlist attempt to set it (E-93). A multi-dimensional vectored port (`in[0:2][0:1]`) is not supported in either declaration order.

Signal-flow disciplines (potential-only or flow-only) are fully usable, including probe-only branches — probing a branch that is never contributed to reads its true flow (an ideal ammeter) instead of the 0-and-open-circuit a naive topology gives.

Discipline compatibility implements all of LRM 3.11.1 (natures audit): a branch — declared or unnamed — between a conservative net and a signal-flow net of the same potential nature is legal (the LRM's own worked example declares `electrical` and `sig_flow_v` compatible: "the nature for flow does not exist in `sig_flow_v`"), a natureless discipline is compatible with its whole domain, and a domainless one with everything. The previous rule demanded every nature binding both-present-or-both-absent, rejecting exactly those pairings — and the diagnostic's help text stated a units-only rule that was never the LRM's. A mixed branch takes the discipline that actually has the natures, so `I(br)` across `electrical`/voltage works. Genuinely incompatible pairings (`electrical` vs `rotational`) are still rejected.

Nature attribute forms are validated again (the checks existed but iterated the wrong accessor and never ran): `access = "SA"` or `access = 3.0` (LRM 3.6.1.2 requires a raw identifier) and a `ddt_nature/idt_nature` value that is not a nature name are located errors at the declaration, instead of being silently dropped — which used to leave the nature with no access function and only a baffling "not found in the current scope" error at the first use. Two more 3.6.1.2 rules are warnings: a *derived* nature declaring units (the declared value is ignored — the parent's units always win — so saying nothing hid a modelling error), and an `idt_nature/ddt_nature` override unrelated (no shared base nature) to the link the parent uses.

Two deliberate deviations stand, now recorded here: $\triangle$ a base nature may omit the `abstol/access/units` attributes 3.6.1.2 requires (E-422's pinned decision — omission produces no wrong answer), and $\triangle$ a derived nature may declare its own access function (E-39 supports it on purpose: `nature Current2 : Current; access = J;` gives a working `J()` — the LRM calls this illegal). Two constructs are not supported and rejected with located diagnostics: vector branches (LRM 3.12's branch `(a,b) br1;` over whole vector nets — use per-bit branches `branch (a[i],b[i])`), and out-of-module discipline declarations (LRM 3.10's `electrical top.other.net;` — meaningless in a single-module OSDI compile).

The shipped **constants.vams** is the VAMS-2023 Annex D.2 file: defining ``PHYSICAL_CONSTANTS_NIST2018` before the include selects the exact 2019-SI values (`P_Q = 1.602176634e-19`, `P_K = 1.380649e-23`, `P_H`, `P_EPS0`, and the measured `P_U0 = 1.25663706212e-6`); the default without the define stays NIST1998 for backward compatibility, exactly as the 2023 LRM specifies. (The 2.4.0 file shipped before had no NIST2018 set, so opting in silently produced NIST1998 values.) The OSDI 0.4 nature descriptors are also exact now: every nature's `num_attr` claimed one attribute more than it owned (a consumer walking the range read the next nature's first attribute), verified fixed by a dlopen dump of the emitted `OSDI_NATURES` table.

Named port branches (LRM 3.7.2) work as of E-84 and carry the same defining equation as a direct port-

flow probe; contributing to one is a proper diagnostic (port branches are probe-only per the LRM):

```
branch (<p>) probe_p;           // named branch over port p
iprobe = I(probe_p);           // same value as I(<p>)
```

Hierarchical branch probes (E-86) read another instance's branches from outside it — named, unnamed, and port-branch forms, the last via a 0V ammeter synthesized at flattening so it reads that instance's own current rather than the node total:

```
V(top.a1.b)                    // named branch of instance a1
V(top.d1.branch(va, vb))       // unnamed branch of instance d1
I(top.d1.branch(<p>))           // d1's own current into its port p
```

4. Expressions (LRM 4)

4.1 Operators and precedence (LRM 4.1, Table 4-2) — ✓

The full operator surface — arithmetic (incl. `**`), comparison, logical, bitwise, shifts, ternary (including over strings), concatenation `{a,b}` and replication `{n{x}}` — was audited operator-by-operator and level-by-level against Table 4-2 with self-checking bitmask modules whose failure modes are numerically visible. Two real defects found and fixed in the audits: unary `~` emitted arithmetic negate, and the constant folder disagreed with the runtime on `>>`; one precedence defect: `%` bound tighter than `*/` (so `6*7%4` evaluated to 18 instead of 2). Associativity verified including `2**3**2 = 64` (left-associative per Verilog) and unary binding above `**` (`-2**2 = 4`). String comparison covers both equality (`==/!=`) and the relational operators (`</<=/>/>=`), the latter a lexicographic (`strcmp`) comparison (E-106).

The arithmetic shifts `<<</>>>` are a flagged extension: LRM 4.2.11 bars them from analog blocks, so every use draws a warning naming the rule — but they keep working, because `>>>` is Verilog's only spelling of a sign-extending right shift (`<<<` is identical to `<<`). $\triangle$ The case (in)equality operators `===/!==` (LRM 4.2.6) lex and evaluate with `==/!=` semantics — exact in a 2-state analog world, which has no x/z bits for them to distinguish (they previously died with a generic parse error that never named the operator). A compile-time shift distance outside `0..=31` is a warning plus the LRM-defined value — the distance is unsigned (LRM 4.2.11), so `1<<32` is 0 and `-8>>>34` is the sign fill, exactly what a runtime distance computes; it used to be a hard error, internally inconsistent with that run-time path. And integer `/` and `%` are now fully defined at the code-generation level: divide-by-zero yields 0 and `INT_MIN/-1` wraps to `INT_MIN`, where LLVM's raw `sdiv/ srem` were undefined behaviour that could SIGFPE an x86 host the moment a runtime divisor crossed zero.

Same-node branch access is an error (LRM 4.4, Table 4-16): `V(a,a)` and `I(a,a)` no longer compile — “the operands of an expression shall be unique to define a valid branch”, and the table marks both forms Error. `V(a,a)` used to compile with no diagnostic at all and silently read 0. A *named* degenerate branch declaration (`branch (a,a) b;`) keeps its E-414 warning: declaring one is survivable, accessing it is not.

`%` by a deck-supplied zero is a run-time error (LRM 4.2.4: “It shall be an error to pass zero (0) as the second argument to the modulus operator”): a literal zero divisor was already a compile error; a zero arriving through a model card now aborts the analysis with `OSDI(fatal) ... %:` the second operand (the modulus divisor) is zero instead of a silent NaN and a generic convergence failure — the same three-route policy as the math builtins (E-509). A genuinely runtime divisor is deliberately left unguarded and evaluates to the defined 0.

4.2 Built-in and environment functions (LRM 4.3, 9.7) — ✓

`ln/log/exp/sqrt/pow/min/max/abs/floor/ceil`, trigonometric and hyperbolic families (integer `min/max/abs` correctly integer-typed; integer division truncates toward zero and real→integer conversion rounds to nearest, ties away from zero, per the LRM — the two are distinct rules; `ceil` of a runtime argument fixed, E-103), `$clog2` returning `ceil(log2 n)` (E-101), the conversion functions

\$rtoi (real→integer, truncating toward zero — as distinct from the rounding implicit cast) and \$itor (integer→real, E-104), and the environment: \$temperature (kelvin — verified as the exact kT/q law across a $-50\ldots150$ °C sweep), \$vt and \$vt(T), \$abstime, \$realttime (identical in the continuous-time analog context), \$simparam and \$simparam\$str, \$param_given, \$port_connected, \$mfactor and the position hidden parameters.

ln1p and expm1 (LRM Table 4-14 and Annex A.8.2) are present in both the bare and \$-prefixed spellings, and lower to libm's log1p/expm1 rather than to $\ln(1+x)/\exp(x)-1$ — the precision near zero is the entire reason the standard lists them separately, and ln1p($1e-18$) returns $1e-18$ where the naive identity returns 0. They were added by E-458 and made *usable* only by E-510: the code generator asked for those two libm routines by name and the intrinsic registry declared neither, so every call whose argument was not constant-folded away crashed the compiler. The gap survived because the suite that introduced them tests \$ln1p(0.5) — a literal, which folds before code generation, so the intrinsic is never emitted. lrmntrin now exercises both spellings with a parameter and with a probe.

Domain errors are refused, by whichever route the value arrives. An out-of-domain *constant* — sqrt(-4), ln(0), asin(2), acosh(0.5), atanh(1), pow with a negative base and a fractional exponent — is a compile-time error naming the builtin, the value and the domain (E-455/479/489/508). The identical value supplied from a model card is an overridable parameter, which the compiler deliberately does not fold, and it used to reach the maths library untouched and return nan/inf silently; E-509 refuses it at run time with the same information. A genuine *run-time* argument is deliberately left alone: sqrt(V(p,n)) legitimately sees a negative argument during Newton iteration, and refusing that would break working models. $\triangle$ That last route is a documented deviation from LRM 4.3.2's unqualified "input values outside of the valid range for the operator shall report an error". One scoping note on the claim above: 0**0 and pow(0,0) evaluate to 1.0 — the near-universal convention, and IEEE 1364's integer-power table says the same — although Table 4-14's pow domain ($x = 0$ requires $y > 0$) reads the point out; the guards flag 0**negative and a negative base with a fractional exponent, not 0**0.

4.3 Analog operators (LRM 4.5) — ✓ (one documented deviation)

Every analog operator produces exact Jacobian contributions via autodiff. The full argument surface (trailing tolerances, optional arguments) was audited at runtime — compiling proves nothing about whether a trailing argument is honored.

```
// time derivative & integrals
I(p,n) <+ ddt(c * V(p,n));
phi = idtmod(K * V(vco), 0.0, 1.0);           // phase wrapping (VCO idiom)
x    = idt(V(in), 0.5, rst);                   // reset form holds at ic

// delays and filters
V(out) <+ absdelay(V(in), 1n);
V(out) <+ transition(sel ? 1.0 : 0.0, 0, 10n, 5n);
V(out) <+ slew(V(in), 1e6, -1e6);              // LRM negative max_neg_rate
V(out) <+ laplace_np(V(in), '{1.0}', '{-6.283e3, 0.0}'); // complex pairs
V(out) <+ zi_nd(V(in), '{0.5, 0.5}', '{1.0}', 1u);

// small-signal & events
gm = ddx(Id, V(g,s));                         // symbolic derivative
tc = last_crossing(V(in) - 0.5, +1);
V(out) <+ ac_stim("ac", 1.0, 90.0);           // module AS the AC stimulus
```

- ddt, idt (with initial conditions that survive into transient, and the assert/reset forms stable enough for relaxation oscillators), idtmod (integrates unbounded internally, wraps the returned value), ddx with respect to potentials *and* flows: ✓

- `absdelay(+maxdelay)`, `transition(3/4/5-argument, `default_transition defaults, DC well-posed)`, `slew` (honoring the LRM's *negative* `max_neg_slew_rate` — the sign defect that made it ignore its input was found in the audit): ✓
- `laplace_nd/np/zd/zp` via exact state-space realization and `zi_nd/np/zd/zp` via bilinear transform, both with complex pole/zero pairs per the LRM's (re, im) vector convention and parameter-dependent coefficients: ✓

Verified against closed-form $H(s)$: all four `laplace_*` forms agree with the analytic one-pole response to $4.7\text{e-}16$ over four decades, a complex pair ($\omega_n = 1/\tau$, $Q = 1$) written as `zp`, `np` and `nd` coefficients agrees to $4.6\text{e-}15$, and all four `zi_*` forms agree with the bilinear $H(z^{-1})$ to $1\text{e-}15$ out to $\omega T = 1$.

A complex root must be written together with its conjugate — the model's duty, not the compiler's. LRM 4.5.11.1 states it directly:

If a root is real, the imaginary part shall be specified as zero (0). If a root is complex, its conjugate shall also be present.

The roots are expanded as $\prod(1 - s/r)$ exactly as the clause's formula shows (including its exception that a root of zero contributes s rather than $1 - s/r$), and the coefficients' imaginary parts cancel *because* the conjugate is there. Writing one root of a pair is therefore a model error, and it does not fail loudly: the imaginary parts no longer cancel, the real part is taken, and a different, physically real filter is built. A $Q = 1$ pole pair written as the single root $\{-0.5/\tau, 0.866/\tau\}$ becomes one real pole at $2/\tau$ — a 20 kHz first-order rolloff in place of a 10 kHz resonant pair, with no diagnostic.

△ Open: that is a detectable mistake for constant roots, which is how they are almost always written, and nothing currently checks it. Recorded here rather than fixed.

- `last_crossing` with simulator-side waveform history: ✓ — and, per LRM 4.5.10, negative until the expression has actually crossed (E-514); it returned `0.0`, which a model testing `last_crossing(...)` < 0 for *not yet* read as a crossing at $t = 0$. For an expression already above zero the zero-seeded crossing history also faked a crossing at the first accepted point, so both the sentinel and the history are now seeded from the operating point.

Analog-operator state at the start of a transient (E-514)

LRM 4.5.7 defines `absdelay` as `Output(t) = Input(max(t - td, 0))`, so before one delay has elapsed the output is `Input(0)` — the operating point, not zero. LRM 4.5.9 says `slew` “returns the value of *expr*” whenever the input is not moving faster than the rate limit, which a constant input never is. Both were violated the same way: the transient's state arrays were seeded with zero, so `slew/transition` ramped up from 0 at exactly the rate limit and `absdelay` read 0 for its whole first delay before stepping to the bias. Both now seed from the converged operating point. LRM 4.5.15's “no state history prior to time $t == 0$ ” is what `max(·, 0)` accommodates; it does not make the value zero.

△ Deviation, deliberate. LRM 4.5.7 also says that with no `maxdelay`, “the value of *td* when the *absdelay*() is first evaluated shall be used and any future changes to *td* shall be ignored”. A time-varying `td` is still tracked. The only first-evaluation flag a model can observe is `IsInitialStep`, and at that flag the circuit solution is still the zero initial guess — a probe read inside `@(initial_step)` returns 0 — so latching there would store 0, which is worse than tracking; and where `td` is a parameter, freezing and not freezing are indistinguishable. Fixing it properly requires the simulator to latch at `MODEINITTRAN`, i.e. a per-slot flag in `0sdiAbsDelayInfo`.

- noise sources (LRM 4.6): `white_noise`, `flicker_noise`, `noise_table`, `noise_table_log` (inline or file data); operating-point-dependent factors and `ddt()`-shaped noise are exact with no extra matrix unknowns; same-named sources sum coherently per LRM 4.6.4 including anti-phase

cancellation: ✓. `noise_table` interpolates the power piecewise-linearly over frequency (LRM 4.6.4.3) and `noise_table_log` interpolates log-log ($P = 10^{(\text{lerp of } \log_{10} p \text{ over } \log_{10} f)}$, LRM 4.6.4.4) — both taking Hz input and clamping to the endpoint powers outside the tabulated range (Enhancement-109 corrected these; they had been a lin-log hybrid and a mis-keyed log-frequency input respectively): ✓

```
I(a,b) <+ white_noise(4.0 * `P_K * $temperature / r, "thermal");
I(d,s) <+ gm * white_noise(gamma_4kT, "induced"); // op-dependent factor
```

- `analysis("ac","tran",...)` variadic, with the AC/noise operating point counting as its parent analysis per LRM 4.6.1: ✓
- `$limit` (built-in "pnjlim" and user functions — genuinely engages SPICE-style iteration limiting), `$bound_step`, `$discontinuity(n)` (clamps the next step *and* makes the integrator bisect onto the event): ✓
- `$table_model` (LRM 9.19 by clause, listed here as a modeling operator): 1-D piecewise-linear, N-dimensional multilinear, and natural cubic-spline interpolation — lowered to differentiable MIR so *all* partial derivatives feed the Jacobian (a table-based MOSFET gets exact gm and gds):

```
I(d, s) <+ $table_model(V(g, s), V(d, s), "mos_iv.tbl", "1L");
```

- $\triangle$ **limexp** is implemented stateless (exact exp below the overflow threshold, tangent-continued above): a stateful previous-iterate limiting version was built and *reverted* because correct converged values under limiting require the simulator's limiting-RHS correction, which applies to circuit unknowns, not to `limexp`'s derived argument. `$limit` provides genuine iteration limiting. Documented decision with the failing evidence preserved.
- **transition** and **slew** reach their final value at `delay + trise`, at every speed — fixed in E-512, and worth recording here because this section previously carried it as a deviation.

Both are one rate-limited tracking loop, $dy/dt = \text{clamp}(K \cdot (x - y), -1/t_{\text{fall}}, +1/t_{\text{rise}})$. While the clamp is saturated that is an exact linear ramp at the LRM's rate; it releases once the remaining gap falls below rate/K , leaving a first-order tail with $\tau = 1/K$. K was a fixed $1\text{e}9 \text{ s}^{-1}$, so the released gap was $1/(K \cdot t_{\text{rise}})$ and the linear fraction depended on how fast the edge was: at `trise = 3 ns` only 66.7% of the swing was a ramp and the output reached 0.8774 instead of 1.0, while at $3 \mu\text{s}$ it was correct. Refining the timestep $100\times$ converged to 0.8776 — to the *wrong* value.

K is proportional to the transition rate now, so the released gap is a fixed fraction at every speed, the endpoint error is $2\text{e-}5 \dots 4\text{e-}4$ across five decades of rise time, and the settled value is exactly 1.0. `transedge` spans those five decades and checks the quarter points of the ramp; `defaulttransition` pinned only the $1 \mu\text{s}$ case, which is why the deviation went unseen for so long.

In DC the filter remains the LRM's static identity $y = x$, selected through the integration-enable parameter — that is what makes the operating point well-posed, and it is independent of the transient shape (E-47).

Out-of-domain operator arguments from the deck (E-504/E-506). The same literal-versus-deck split described for the built-in functions in §4.2 applies to the operators' own numeric arguments, and each case is resolved by what the argument *means* rather than by a blanket rule:

- a `zi_*` sampling period ≤ 0 inverts the bilinear map and reflects every pole across the imaginary axis; there is no honest value to project it onto, so the run time says what the compiler says and aborts, naming the value;
- a `laplace_*` leading denominator coefficient of zero is a division by zero and aborts the same way;
- an `idtmmod` modulus ≤ 0 falls back to the unwrapped integral, because that is exactly what `idtmmod` means with no modulus supplied — the model keeps running and the value stays finite;

- the `$rdist_*` family clamps onto the nearest distribution it can actually produce (a negative standard deviation behaves as 0), rather than returning the exact negation of the valid deviate as it once did.

Restrictions enforced as the LRM demands (4.5.1): analog operators inside loops or solution-dependent conditionals are rejected with diagnostics that name the actual construct and cite the clause. The conditional rule is applied as the LRM states it — only a condition that depends on an analog signal is refused, so the common idiom of gating an operator on a parameter (`if (SELFHEATING) ... ddt(...)`) compiles.

5. Analog behavior (LRM 5)

5.1 Analog blocks (LRM 5.2, 6.2) — ✓

Multiple analog and `analog initial` blocks per module execute as-if-concatenated in source order — verified including across instance flattening and with declarations interleaved between blocks.

5.2 Contribution statements (LRM 5.6) — ✓

Direct contributions with accumulation semantics, switch branches (the two kinds on mutually exclusive conditional paths — contributing both with no condition between them keeps only the last, which E-400 reports), indirect branch assignment (the LRM's ideal-op-amp construct), and port-flow probes:

```
V(out): V(inp, inn) == 0;           // indirect: drive out so V(inp,inn)=0
I(p,n) <+ V(p,n)/r;                // accumulates with other <+ statements
iport = I(<p>);                    // total flow through port p (ideal probe)
branch (<p>) pb;                  // named port branch: I(pb) == I(<p>) (E-84)
```

`$port_connected` works on connected *and* unconnected ports of flattened instances (resolved per instance at elaboration time), and instantiating an undefined module is a hard error rather than a silent drop (both E-84). Bus actuals may be part-selects — positional, named, or width-1 onto a scalar port (E-85):

```
adc2 hi (out[3:2], in);           // positional slice
adc2 lo (.out(out[1:0]), .in(r)); // named slice
```

5.3 Procedural statements (LRM 5.7–5.9, 5.11) — ✓

`if/else`; `case` over integers, reals, strings, and arrays (element-wise); all four loops with runtime trip counts that honor model-card parameter overrides; `disable` as the LRM's named-block early exit; and the VAMS-2023 jump statements `break/continue/return` (LRM 5.11, Annex-C audit): `break/continue` act on the innermost runtime loop of any kind (`for`'s `continue` re-enters at the increment, `repeat`'s still counts the iteration), `return [expr]` exits an analog function — from arbitrarily nested statements — after optionally setting its return value:

```
for (i = 0; i < n; i = i + 1) begin
  if (w[i] == 0.0) continue;           // skip, increment still runs
  if (i == cutoff) break;
  acc = acc + w[i] * x[i];
end
```

```
analog function real findk;
  input lim; integer lim; integer k;
  begin
    findk = -1.0;
    for (k = 0; k < 10; k = k + 1)
```

```

        if (k == lim) return k;    // early exit, value set
    end
endfunction

```

Position rules are enforced: break/continue outside a runtime loop — including inside a genvar analog_for, which LRM 5.9.3 excludes from jump statements (those loops unroll at elaboration) — and return outside an analog function are targeted errors. The keywords are contextual: pre-2023 source that used break or return as an identifier still compiles, flagged by the L012 keyword-compat lint like the other VAMS reserved words.

5.4 casex / casez — $\triangle$ extension beyond Annex C

The don't-care case variants are legal in full Verilog-AMS but Annex C.7 excludes them from Verilog-A ("casex and casez are not supported in Verilog-A"); this compiler supports them anyway as a deliberate superset extension, with 2-state semantics: x/z/? digits of a based literal written directly as an item form a comparison mask — the arm matches on every *care* bit. casex treats x/z/? as don't-cares, casez only z/?. The classic priority-encoder idiom (from the committed [examples/casexz_examples/](#)):

```

casex (req)
    4'b1xxx: grant = 8;    // highest set bit wins
    4'b01xx: grant = 4;
    4'b001x: grant = 2;
    4'b0001: grant = 1;
    default: grant = 0;
endcase

```

Don't-care literals anywhere else, x digits under casez, and non-integer discriminants are located errors — no silent zeros.

5.5 Events (LRM 5.10) — ✓

@(initial_step) and @(final_step) genuinely gate (once at the start; once at the *successful* end, seeing the converged solution), with analysis-phase lists filtering by analysis type; cross/above/ timer with tolerances and persistent state; event OR lists:

```

@(initial_step("ac", "tran")) voff = 0.1;
@(cross(V(s) - vth, +1) or timer(0, 1u))
    n_events = n_events + 1;
@(final_step) $strobe("peak = %g at end of run", peak);

```

An operating point fires both step events (a single point is first and last); a failed analysis never fires final_step.

5.6 Analog functions (LRM 5.11) — ✓

Declared functions with input/output/inout arguments, arrays in all three roles plus array return values, locals (including array locals), loops inside bodies; inlined at the call site so derivatives flow through automatically:

```

analog function real dot;
    input real a[0:3], b[0:3];
    integer k;
    begin
        dot = 0;
        for (k = 0; k <= 3; k = k + 1)
            dot = dot + a[k]*b[k];
    end
endfunction

```

```
end
endfunction
```

Recursion is illegal in Verilog-A; both direct and mutual recursion are clean errors naming the call cycle (mutual recursion formerly overflowed the compiler stack).

6. Hierarchy (LRM 6) — ✓

Module instantiation with named and positional ports and parameter overrides, open ports, instance arrays, arbitrary nesting, across `include boundaries; **generate** in all three forms with genvars usable in any expression; **defparam** with its LRM precedence over instance overrides; hierarchical names and \$root; implicit nets; net concatenation in port connections:

```
// genvars usable in any expression, e.g. a parameter override
module gpexpr(a, c);
    inout a, c; electrical a, c;
    genvar i;
    generate
        for (i = 0; i < 3; i = i + 1) begin : g
            res #(.r(1e3*(i+1))) ri (a, c);    // 1k || 2k || 3k
        end
    endgenerate
endmodule

defparam u1.r = 2e3;                // hierarchical override
defparam u1.u2.r = 4e3;            // two levels down; wins over #(...)
```

One boundary the OSDI target makes explicit (and the compiler explains in its diagnostic): structure binds at compile time — a generate condition or bound may depend on genvars and literals but not on module *parameters*, because parameters arrive from model cards at simulation time, after structure is frozen. Loop trip counts and if arms, being behavior, may depend on parameters freely. [△](#) documented deviation with rationale.

7. System tasks and functions (LRM 9)

7.1 Display (LRM 9.4) — ✓

All five kinds (\$strobe, \$display, \$write, \$monitor, \$debug) with the complete format surface — [flags] [width] [.precision] on every conversion, dynamic * widths, %e/f/g/r (engineering notation), %d/h/o/b/c/s, %m, %% — verified against exact printf output. As of [E-516](#) the LRM 9.4.6 timing rule holds: display output is buffered per Newton iteration and printed only for the accepted point (\$debug keeps the clause's exemption and still prints per iteration; statements inside event-controlled blocks print immediately, since they fire on the event's own iteration). \$monitor implements 9.4.1's change detection — one line per change of its argument list at accepted points (a \$abstime argument defeats the text comparison — documented). %r engineering notation, which printed garbage for every input, is exact: 1e3 → 1.000000k, 1e-9 → 1.000000n, 0.036 → 36.000000m. A null argument (two adjacent commas) renders as the single space 9.4.1 specifies. A constant-argument display hoisted to instance initialization still executes there ([△](#) documented at the split):

```
$strobe("Vth=%7.4f V  region=%2d  id=%r  name=%s", vth, reg, id, name);
```

7.2 File and string I/O (LRM 9.5, 9.8) — ✓

\$fopen/\$fclose/\$fdisplay/\$fwrite/\$fstrobe/\$fmonitor/\$fdebug/\$fflush/ \$ftell/
\$fseek/\$rewind/\$feof/\$ferror/\$fgetc/\$ungetc (single-character read and one-character
pushback, E-107/E-108) and \$swrite/\$sformat/\$sscanf/ \$fgets/\$fscanf. The 9.5 lifecycle

rules hold as of E-516: un-gated file writes are deferred to the accepted iteration (9.5.9, with readable streams rewind to their accepted baseline each iteration for the read replay); a "w"-mode reopen in a following analysis appends (9.5.1.1); the open-write-close idiom in the analog body writes its line (an instance-setup `$fclose` defers so the descriptor survives for eval); a re-run of the instance initialization reproduces `$rewind/$fseek` files byte-exactly; and the pre-opened descriptors 32'h8000_0000/1/2 reach stdin/stdout/stderr. Full one-hot multichannel descriptors and `$fmonitor` change detection remain $\triangle$ documented gaps. `$sscanf/$fscanf` honour the format conversion base — %h/%x hex, %o octal, %b binary, %d decimal (E-105).

The scan format is read only when it is a literal (E-507). The scanners are chosen at compile time from the format text, so a format that is not a literal cannot select them: such a call used to read the *text of the format itself* as data and report success. It is refused now, as every other builtin needing a compile-time string already did. Within a literal format, the parts that the scanners cannot honour are refused with a reason rather than silently ignored — literal characters between conversions, a field width, %* assignment suppression (which returned the very field it asks to discard), and unsupported conversion characters. A conversion that does not match leaves its destination unchanged rather than zeroing it, so a scan into a variable that already holds a value is non-destructive:

```
integer fd;
analog @(final_step) begin
    fd = $fopen("iv.dat");
    $fdisplay(fd, "%g %g", vpeak, ipeak);
    $fclose(fd);
end
```

7.3 Random and distributions (LRM 9.13) — ✓ (deterministic semantics)

`$random`, `$arandom`, and every `$dist_*`/`$rdist_*` (uniform, normal, exponential, poisson, chi-square, t, erlang), verified against closed-form moments. $\triangle$ One deliberate semantic: draws are a deterministic function of (*seed*, *call site*) with no in-place seed advance — an advancing seed would return different values on every Newton iteration and destroy convergence. This gives reproducible Monte Carlo and independent per-instance variation, at the cost of in-evaluation sequences (which have no convergent meaning in an analog solver anyway).

Return types follow the LRM split: `$dist_*` returns integer and `$rdist_*` returns real — the `$rdist_*` family exists precisely because the originals, inherited from Verilog-2001 §17.9.2, are integer-only. Both returned real until E-376, which made the LRM-conformant spelling `$display("%d", $dist_uniform(s, 10, 20))` compile; it had been rejected with *"expected integer value but found real value"*. The draws themselves did not move — only the static type — and the discrete/continuous split is visible in the moments: `$dist_uniform` sits at $(n^2-1)/12$ while `$rdist_uniform` sits at $100/12$.

```
parameter integer seed = 1;
real gain;
analog begin
    gain = 1.0 + sigma * $rdist_normal(seed, 0.0, 1.0); // per-instance mismatch
    ...
end
```

7.4 Simulation control (LRM 9.6–9.7) — ✓

`$finish`, `$stop`, `$fatal` are honored with simulator-correct timing (at the accepted-point boundary; `$finish` fires `@(final_step)` and ends the analysis cleanly; `$fatal` during setup reads as a configuration rejection), and `$warning`/`$error`/`$info` display.

7.5 The mechanism-unavailable group —

`$simprobe`, `$analog_node_alias/$analog_port_alias` compile and return their LRM-specified fallback values (supplied default / 0). (`$test$plusargs/$value$plusargs` are NO LONGER in this group: [Enhancement-215](#) serves command-line plusargs through the `simparam` channel.) This is the LRM's own escape hatch for hosts without the mechanism; models using them run predictably instead of being rejected.

7.6 Attributes as simulator metadata — ✓

(`* desc="..." *`) on a module-scope variable exposes it as an operating-point variable (`print @n1[gm]`, `.save`, `.meas`) — a described variable declared inside a named block is local to that block (LRM 3.2.1) and is no longer exported as a phantom op-var; (`* type="instance" *`) on a parameter puts it on the instance line and under `alter/.dc` sweeps; `desc/units` on parameters populate the OSDI descriptor.

8. Compiler directives (LRM 10) — ✓

The predefined source-location macros work (E-85): `__FILE__` expands to the source file's basename, `__LINE__` to the exact line (a use inside a ``define` body reports the definition site — documented textual-expansion semantics):

```
$strobe("evaluated at %s:%0d", __FILE__, __LINE__);
```

``define` with arguments (macros-in-macros, macro calls as arguments), ``ifdef`/`ifndef`/`elsif`/`else`/`endif` chained and nested, ``include` chains, ``undef` (which, per LRM 10.4, has no effect on the predefined macros and warns — [E-515](#), which also makes `__VAMS_ENABLE__` always-defined per 10.5, validates ``begin_keywords`/`end_keywords` version specifiers per 10.6, and fixes `__FILE__`/`__LINE__` breaking relative ``include` resolution), ``resetall` (now genuinely resetting the tracked directive state), ``default_transition`, and the housekeeping set (``celldefine`, ``timescale`, ``line`, ``pragma`, ``default_nettype`, ...). Recursive macro expansion — direct or mutual — is a clean located error (formerly a compiler stack overflow), while the legal same-macro nesting inside *arguments* (``QUAD(x) = `TWICE(`TWICE(x))`) expands finitely:

```
`define TWICE(x) ((x)*2.0)
`define QUAD(x) (`TWICE(`TWICE(x))) // same-macro nesting in ARGUMENTS: legal
`ifdef HIGH_PRECISION
    `define TOL 1e-12
`else
    `define TOL 1e-9
`endif
```

(*`default_discipline* is AMS-only and out of scope — implicit nets themselves are Verilog-A and work, taking their discipline from the connected port.)

9. Compliance summary

LRM area	Status	Verified by
2 Lexical (identifiers, numbers, strings, attributes, three-token based literals, expression-position attrs)	✓ (E-515 audit)	escid, stresc, comment, lrmlex
3.2–3.3 Value types, variables, persistence	✓	vartype, variable_persistence, intstate, varinit

LRM area	Status	Verified by
3.4 Parameters (ranges, localparam, aliasparam incl. 3.4.7 error rules, arrays incl. whole-array instantiation override, paramset, block-scoped)	✓ (default-exemption △ and frozen-type-of-untyped △ documented)	paramrange, localparam, array, paramarray, paramset, paramsethsp, blockparam, alias
3.5–3.13 Natures, disciplines, nets, buses, nodesets, 3.11.1 compatibility (signal-flow/natureless/domainless), nature-attribute validation, NIST2018 constants	✓ (base-nature attr omission △ and derived-access extension △ documented; vector branches ? rejected)	derivednature, domainbind, bus, netinit, ground, signalflow
4.1–4.4 Operators (incl. string relational, ==/!= as 2-state ==/!=), precedence, functions (\$clog2, \$rtoi/\$itor, ln1p/expm1, domain diagnostics both routes, %-by-deck-zero fatal, same-node V(a,a)/I(a,a) rejected)	✓ (<<</>>> extension △ warned; runtime-probe domain policy △ documented)	operator, precedence, shift, concat, stringcmp, clog2, convert, ceil, lrmfuncs, lrmintrin, domainrt
4.5 Analog operators (all), clause-by-clause audit 4.5.1–4.5.15 (E-514)	✓ (limexp stateless △ documented; absdelay frozen-td △ below)	absdelay, laplace, zi, slew, transition, idt*, ddx, opargs, discontinuity, last_crossing, defaulttransition, transedge, rtdomain, deckdomain
4.6 Noise (incl. correlation 4.6.4; noise_table linear-in-f, noise_table_log log-log)	✓	noise, noisetable, noisecorr, noisejw
5.2/6.2 Analog blocks (multiple)	✓	multianalog
5.6 Contributions, indirect assignment, port flow (incl. named port branches)	✓	indirect_assignment, portflow, signalflow, lrm
5.7–5.9, 5.11 Procedural statements, loops, disable, jump statements (break/continue/return, contextual keywords, genvar-for exclusion enforced)	✓	analogloop, dowhile, repeat, disable, arraycase
casex/casez	△ extension beyond Annex C (C.7 excludes them from Verilog-A)	casexz
5.10 Events (steps, cross/above/timer, OR lists, phase lists)	✓	initial_step, finalstep, cross, timer, lrmcorner
4.7 Analog functions (arrays in/out/return; VAMS-2023 string return & args; return statement)	✓	funcarray, arrayout, arrayret

LRM area	Status	Verified by
6 Hierarchy (instantiation, generate incl. the legacy analog-block form — obsolete per G.2.3/C.20, kept as a compat extension $\triangle$ — defparam, \$root, part-select connections, hierarchical branch probes)	✓ (param-shaped structure $\triangle$ explained)	instantiation, generate, legacygen, defparam, hiername, implicitnet, partselect, hierbranch
9.4–9.8 Display, file/string I/O (incl. 9.4.6/9.5.9 accepted-iteration deferral, \$monitor change detection, %r, 9.5.1.1 append, pre-opened fds)	✓ (E-516 audit; MCD/\$fmonitor $\triangle$)	display, fileio, stringio, fgetc, ungetc, sscanf, scanfmt, lrmsysio
9.13 Random/distributions	✓ (deterministic seed $\triangle$ documented)	rng, montecarlo
9.17.3 / Annex E.3.4 \$limit	✓ (unknown name falls back to no limiting with a warning, per 9.17.3; vdslim = Table E.2's preferred spelling of limvds)	fetlim, lrm
Annex E SPICE primitives (E.2/E.3, Table E.1)	$\square$ not supported: resistor/capacitor/bjt/... cannot be instantiated from Verilog-A source (clean error); instantiate SPICE devices at netlist level instead	—
9 misc (\$finish family, \$simparam, attributes)	✓	simctrl, simparamstr, opvar
plusargs (\$test/\$value)	✓ (E-215)	plusargs
simprobe / node aliases	$\triangle$ LRM fallbacks	alias
10 Compiler directives (incl. `__FILE__`/ `__LINE__`, predefined-macro protection, `begin_keywords`)	✓ (E-515)	preproc, directive, defaulttransition, filemacro, lrmlex
Mixed-signal / digital (outside Annex C)	× by scope	—

As of E-84 the standard's own examples are a compliance suite: every code example in the LRM 2023 PDF is extracted and compiled (examples/lrm_examples/ — per its manifest.json: 47 compile, 12 documented limitations pinned to their diagnostics, 21 correctly rejected as mixed-signal), and the 146 non-module fragments double as a no-crash fuzz corpus.

Two later findings are worth recording here because both were *compliance* failures rather than robustness ones — legal Verilog-A that the compiler refused or mishandled, found only after the construct-level suites above were all green:

- Array parameters could not survive instantiation (E-363). A module declaring parameter `real cf[0:2]` could not be instantiated twice: elaboration renamed scalar parameters and array *variables* per instance but not array *parameters*, so both instances re-declared `cf[0]`, `cf[1]`, ... and name resolution rejected the second. The same collision hit two *different* modules that merely shared an array-parameter name. This sits across LRM 3.4 (parameters) and LRM 6 (hierarchy), and no single-feature test could see it — it needs a parameter form and instantiation together.
- A **case** inside a **do-while** crashed the compiler (E-363) — LRM 5.7/5.9 constructs that are individually covered by the arraycase and dowhile suites, but had never been *composed*.

- Two standard functions crashed the compiler, while passing their own test ([E-510](#)). `ln1p` and `expm1` (LRM Table 4-14) lower to `libm` routines the code generator requests by name, and the intrinsic registry declared neither — so every call whose argument was not constant-folded away aborted the compiler. The suite that introduced them tested `$ln1p(0.5)`, and a literal folds before code generation, so the intrinsic was never emitted and the test could not reach the broken path. This is worth recording as a methodology point rather than a bug: for anything that lowers to a call, a literal argument does not exercise code generation, and a construct-level suite written only with literals can be green over a feature that no user can use. The fix was two declarations; finding it took fuzzing 2 089 constant-folded expressions and noticing that exactly two crashed for *every* argument.

Both of the first two came from a generator that emits valid-by-construction programs combining features developed in isolation, which is a different instrument from the mutation fuzzing used earlier: mutants die at the parser, so they never reach the semantics. One defect from the same round was left open at the time — a provably non-terminating analog loop — and is now closed by [E-375](#): it is a compile-time error. There is no correct object code for a model that cannot complete one evaluation, so a diagnostic was always the only right answer. Worth recording why it could not simply be left: the E-363 CFG repair had stopped the compiler crashing and started it *emitting*, and the resulting `.osdi` loaded cleanly and then hung the simulator on the first device evaluation with no diagnostic at all — strictly worse than the crash.

Separately, the whole 496-file corpus now compiles with zero assertion failures under an assertions-enabled build ([E-347](#)), which is a stronger statement than the release-build corpus run: a release build compiles every `debug_assert!` out, so internal-invariant violations pass silently there.

Beyond construct-level checks, three cross-cutting guards pin that the *whole language implementation* produces correct physics: the 92-model industry corpus compiles and runs, the static/dynamic/thermal physics suites recover textbook device laws through the full pipeline (60 mV/dec, $C \equiv dQ/dV$ across analyses, $E_g \approx 1.25$ eV from a compiled MEXTRAM), and the twin benchmarks show OSDI output matching hand-coded simulator devices to machine precision.